

Figure 1: Max-cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of edges crossing $(S, V \setminus S)$; $G(n, p)$ with integer weights in $\{1, \dots, 9\}$. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+ (T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

Random greedy: certified ratio vs. k , with the B2–B4 baselines

Each plot shows RG/B for every upper bound B : the dual bound, top- k singletons, total weight, the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3), the modular bounds μ_2, μ_3 (B4) and the Marginal bound (the dual bound without its caps), against the $1/e$ guarantee of random greedy.

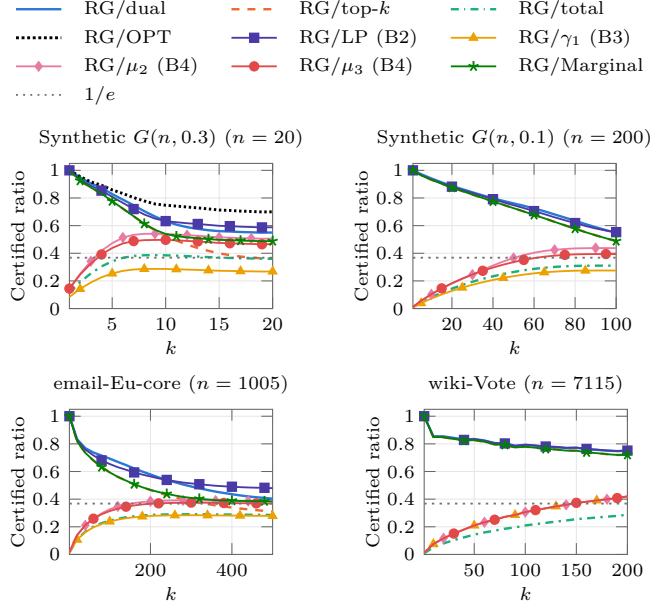


Figure 2: Directed cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of edges leaving S ; synthetic $G(n, p)$ with random orientations and SNAP graphs. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 10th greedy prefix only. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+ (T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

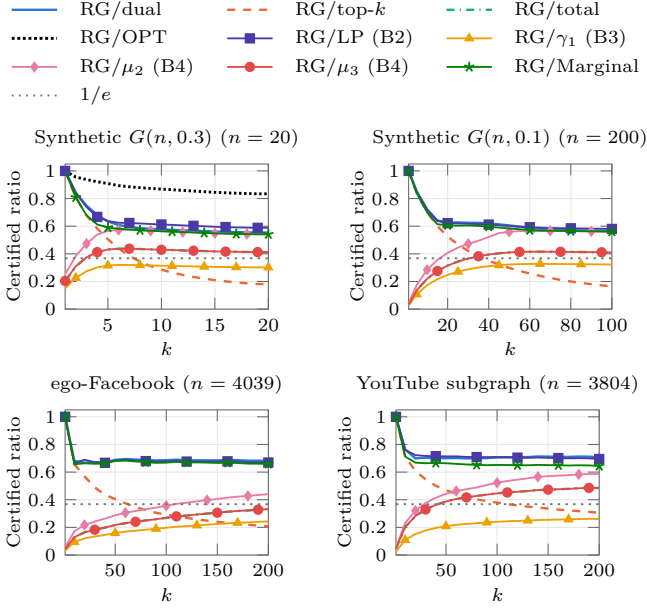


Figure 3: Revenue maximization: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \sum_{i \notin S} \sqrt{\sum_{j \in S} w_{ij}}$ with $w_{ij} \sim U(0, 1)$. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 5th or 10th greedy prefix only. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

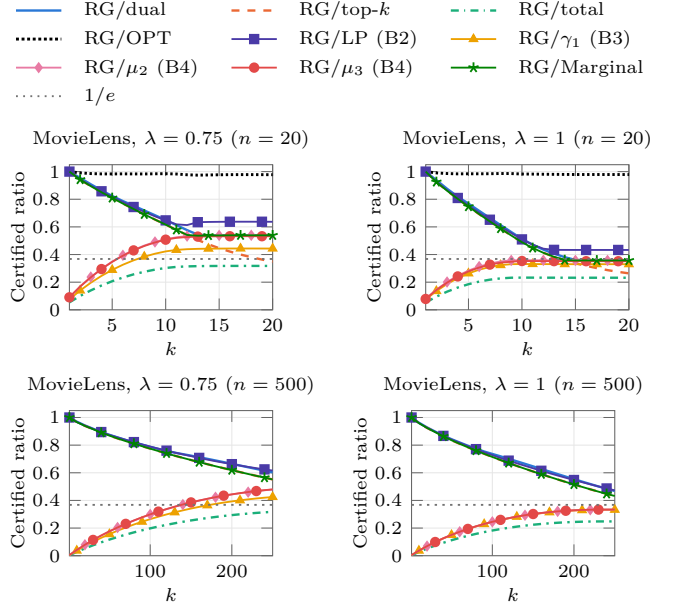


Figure 4: Diverse recommendation: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \sum_{i \in S} \sum_{j \in V} s_{ij} - \lambda \sum_{i,j \in S} s_{ij}$, cosine similarities of MovieLens 1M rating vectors. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

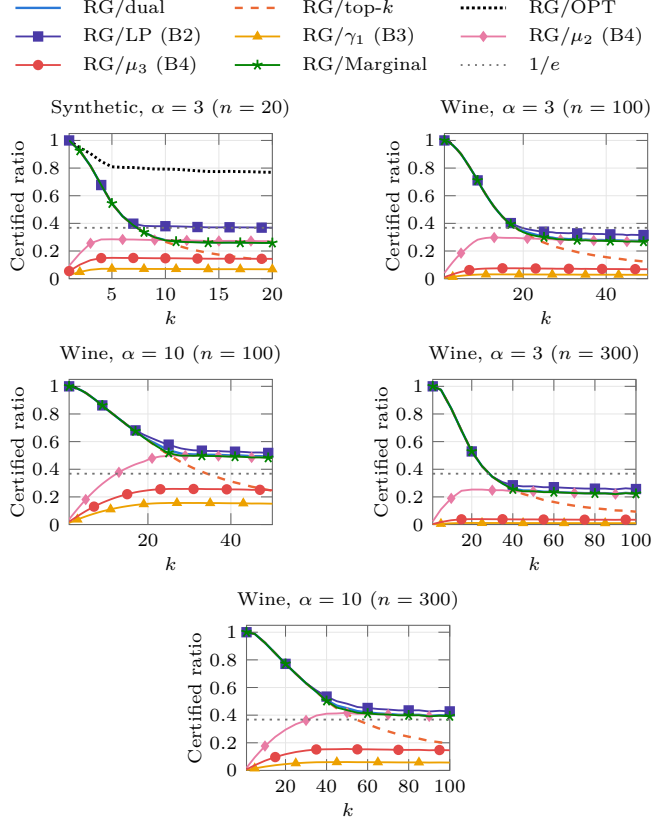


Figure 5: Log-determinant: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = \log \det(L_S)$, $L = \alpha(K + 10^{-3}I)$ with an RBF kernel K on the features. RG/OPT uses the exact optimum (brute force) where it was computable. On the largest instances the dual bound is minimized over every 5th greedy prefix only. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+ (T_k(S))$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

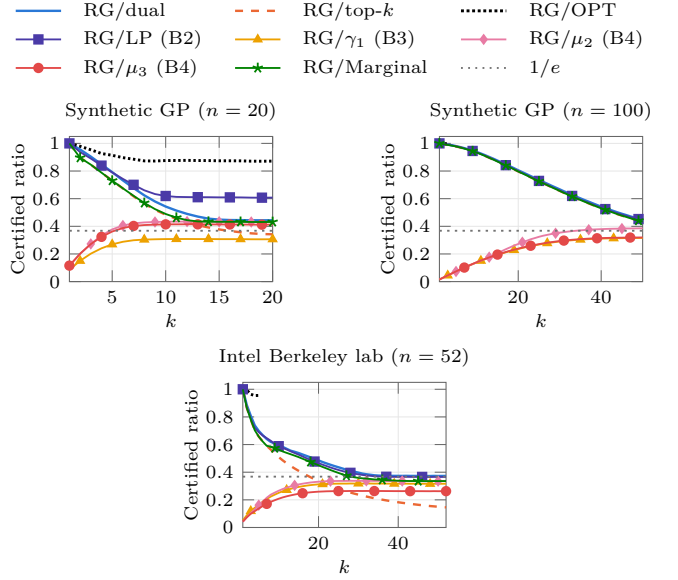


Figure 6: Gaussian mutual information: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S) = I(X_S; X_{V \setminus S})$ for a Gaussian process on random points and for the Intel Berkeley lab temperature sensors. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a | S)]^+ (T_k(S))$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.

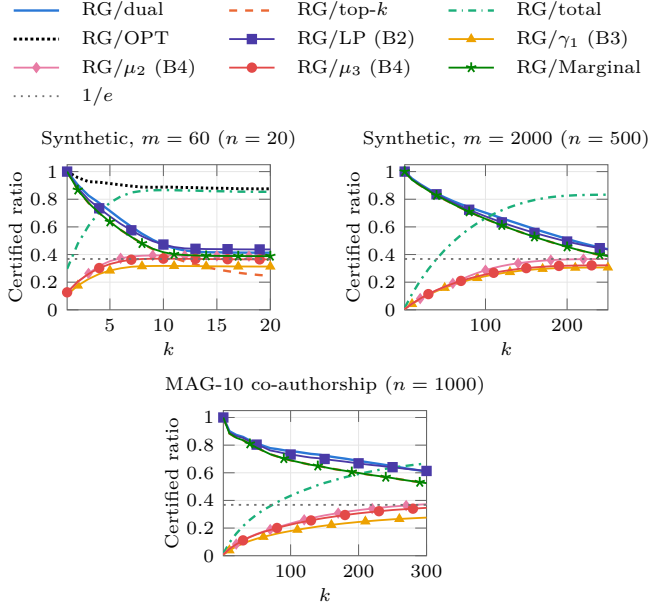


Figure 7: Hypergraph cut: certified approximation ratio of random greedy (RG, mean over trials) with respect to each upper bound, as a function of the cardinality k . $f(S)$ = weight of hyperedges split by S ; random hypergraphs and the MAG-10 co-authorship hypergraph. RG/OPT uses the exact optimum (brute force) where it was computable. Additional baselines: the LP over the dual’s base sets (B2), the pruned double-greedy bound γ_1 (B3) and the modular bounds μ_2, μ_3 on the pruned lattice (B4); B3 and B4 bound the unconstrained optimum. Marginal is $\min_S f(S) + \text{Pen}(S) + \sum_{a \in T_k(S)} [f(a \mid S)]^+$ ($T_k(S)$: the k largest marginals) over the same base sets, i.e. the dual bound without its caps. Curves are means over graph/sample seeds.